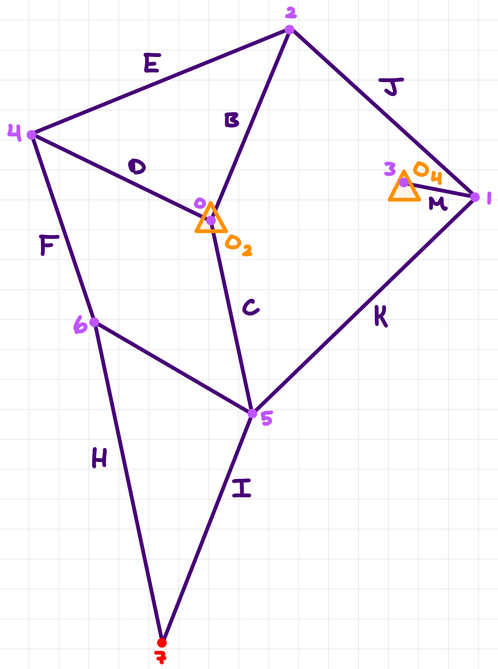
Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Simplified Design for Kinematic Analysis:

The gear can be modeled as a link (M) as it acts as a crank/Input Link.

Each Leg is fixed at 2 points O_2 & O_4.



Dimensions:

- Link B ~ 62.25
- Link C ~ 58.95
- Link D ~ 60.15
- Link E ~ 83.7
- Link F ~ 59.1
- Link G ~ 55.05
- Link H ~ 98.55
- Link I ~ 73.5
- Link J ~ 75.0
- Link K ~ 92.85
- Link M ~ 22.5

Invisible / Hidden Links:

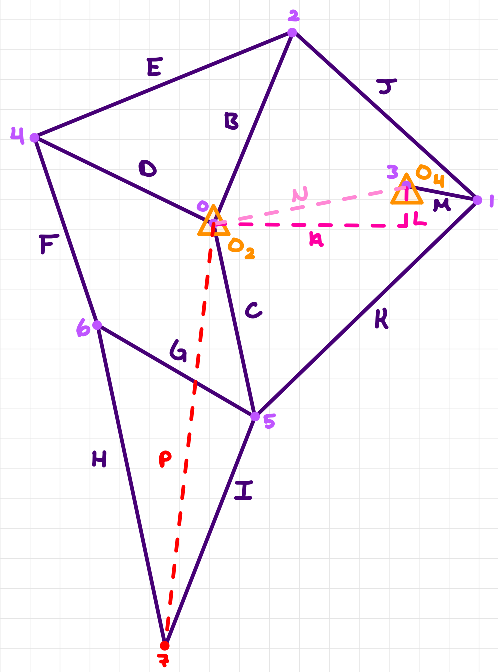
The mechanism has 2 hidden Links (A & L) which offsets the fixed points.

The mechansim also has 2 Invisible Links which are useful in simplifying the mechanism.

Furthermore, by adding the 2 Invisible Links (N & P) it will allow for the System to be Solved by Vector Loop Analysis.

Link N ~ Ground Link

Link P ~ Toe Link



Dimensions:

- Link A ~ 57.0
- Link L ~ 11.7
- Link N ~ $\sqrt{\text{Link A}^2 + \text{Link L}^2} \approx 58.1884$
- Link P ~ Changing

Course.	Title.	Date. Mo Tu We Th Fr Sa Su

Links & Linkages:

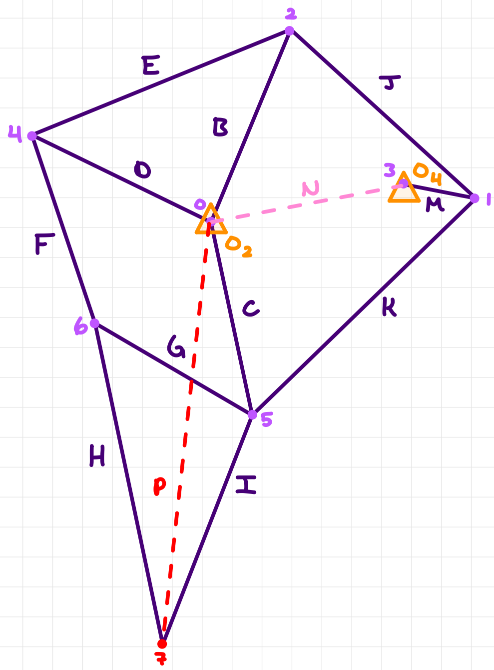
Rigid Bodies:
Links (B, D, & E)
Links (G, H, & I)

Upper Four-Bar Mechanism: (NMBJ)
Link N, Link M
Link B, Link J (Crank-Rocker)

Lower Four-Bar Mechanism: (NMCK)
Link N, Link M
Link C, Link K (Crank-Rocker)

Parallel Mechanism: (DCFG)
Link D, Link C
Link F, Link G (open 4-Bar Linkage)

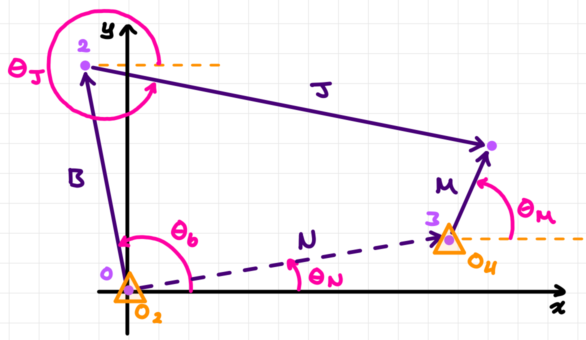
Toe Mechanism: (CIP)
Link C, Link I
Link P,



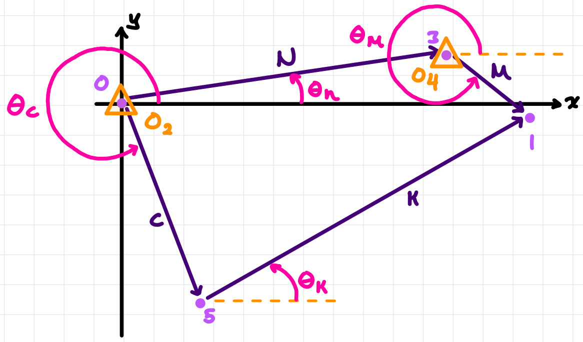
The upper, Lower, Parallel, & Toe mechanism are all Individual Loops & are all 4 or 3 Bar Mechanisms allowing for Vector Loop Analysis of the 4 different Mechanisms

Vector Loops of the Mechanism:

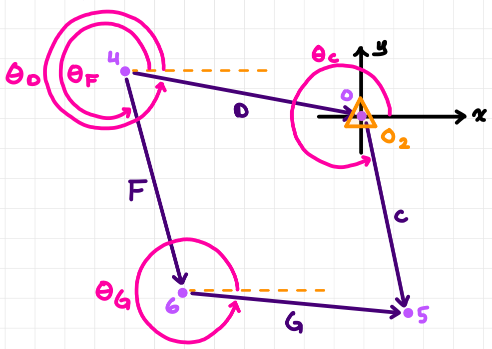
a) NM-BJ



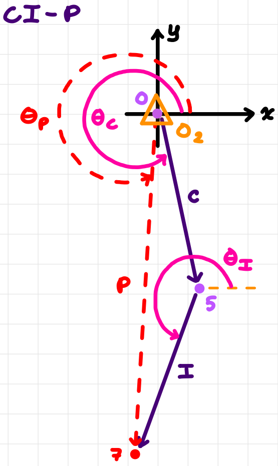
b) NM-CK



c) DC-FG



d) CI-P

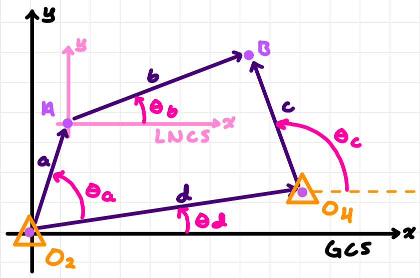


Course.	Title.	Date. Mo Tu We Th Fr Sa Su

General Four-Bar Linkage:

GCS ⤳ General coordinate System

LNCS ⤳ Local Non-Rotating Coordinate System



Position Analysis:

General Vector Loop EQ: (dc-ab)

$$\boxed{\vec{d} + \vec{c} - \vec{a} - \vec{b} = 0}$$

Using Complex Numbers:

$$\boxed{de^{j\theta_d} + ce^{j\theta_c} - ae^{j\theta_a} - be^{j\theta_b} = 0}$$

Where $a, b, c, \& \ d$ are the Link Length

$\theta_c \ \& \ \theta_d$ are Known Angles

$\theta_a \ \& \ \theta_b$ are unknown Parameters

Corresponding Parameters:

Upper Four-Bar Loop: (NM-BJ)

Lengths: $\{a, b, c, d\} \triangleq \{B, J, M, N\}$

Angles: $\{\theta_a, \theta_b, \theta_c, \theta_d\} \triangleq \{\theta_B, \theta_J, \theta_M, \theta_N\}$

$$\boxed{Ne^{j\theta_N} + Me^{j\theta_M} - Be^{j\theta_B} - Je^{j\theta_J} = 0}$$

Lower Four-Bar Loop: (NM-CK)

Lengths: $\{a, b, c, d\} \triangleq \{C, K, M, N\}$

Angles: $\{\theta_a, \theta_b, \theta_c, \theta_d\} \triangleq \{\theta_C, \theta_K, \theta_M, \theta_N\}$

$$\boxed{Ne^{j\theta_N} + Me^{j\theta_M} - Ce^{j\theta_C} - Ke^{j\theta_K} = 0}$$

Parallel Loop: (DC-FG)

Lengths: $\{a, b, c, d\} \triangleq \{F, G, C, D\}$

Angles: $\{\theta_a, \theta_b, \theta_c, \theta_d\} \triangleq \{\theta_F, \theta_G, \theta_C, \theta_D\}$

$$\boxed{De^{j\theta_D} + Ce^{j\theta_C} - Fe^{j\theta_F} - Ge^{j\theta_G} = 0}$$

Vector Addition:

Toe Loop: (CI-P)

$\vec{P} = \vec{C} + \vec{I}$

$$\boxed{Pe^{j\theta_P} = Ce^{j\theta_C} + Ie^{j\theta_I}}$$

Note. The unknown parameters are $P \ \& \ \theta_P$ for Toe Loop.
uses Vector Addition instead of Vector Loop Analysis

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Additional Equations for θ_D & θ_I:

Rigid Body: (BDE)

$\theta_D = \theta_B + \theta_E$

$E^2 = B^2 + D^2 - 2BD\cos\theta_E$

$\theta_E = \cos^{-1}\left(\frac{B^2+D^2-E^2}{2BD}\right)$

$$\boxed{\theta_D = \theta_B + \cos^{-1}\left(\frac{B^2+D^2-E^2}{2BD}\right)}$$

Rigid Body: (GHI)

$\theta_I = \theta_G - \theta_H$

$H^2 = G^2 + I^2 - 2GI\cos\theta_H$

$\theta_H = \cos^{-1}\left(\frac{G^2+I^2-H^2}{2GI}\right)$

$$\boxed{\theta_I = \theta_G - \cos^{-1}\left(\frac{G^2+I^2-H^2}{2GI}\right)}$$

Real & Imaginary EQ:

Upper Four-Bar Loop: (NM-BJ)

$Ne^{j\theta_N} + Me^{j\theta_M} - Be^{j\theta_B} - Je^{j\theta_J} = 0$

Real: $\boxed{N\cos\theta_N + M\cos\theta_M - B\cos\theta_B - J\cos\theta_J = 0}$

Imaginary: $\boxed{N\sin\theta_N + M\sin\theta_M - B\sin\theta_N - J\sin\theta_N = 0}$

Lower Four-Bar Loop: (NM-CK)

$Ne^{j\theta_N} + Me^{j\theta_M} - Ce^{j\theta_C} - Ke^{j\theta_K} = 0$

Real: $\boxed{N\cos\theta_N + M\cos\theta_M - C\cos\theta_C - K\cos\theta_K = 0}$

Imaginary: $\boxed{N\sin\theta_N + M\sin\theta_M - C\sin\theta_C - K\sin\theta_K = 0}$

Parallel Loop: (DC-FG)

$De^{j\theta_D} + Ce^{j\theta_C} - Fe^{j\theta_F} - Ge^{j\theta_G} = 0$

Real: $\boxed{D\cos\theta_D + C\cos\theta_C - F\cos\theta_F - G\cos\theta_G = 0}$

Imaginary: $\boxed{D\sin\theta_D + C\sin\theta_C - F\sin\theta_F - G\sin\theta_G = 0}$

Toe Loop: (CI-P)

$Pe^{j\theta_P} = Ce^{j\theta_C} + Ie^{j\theta_I} \Rightarrow Ce^{j\theta_C} + Ie^{j\theta_I} - Pe^{j\theta_P} = 0$

Real: $\boxed{C\cos\theta_C + I\cos\theta_I - P\cos\theta_P = 0}$

Imaginary: $\boxed{C\sin\theta_C + I\sin\theta_I - P\sin\theta_P = 0}$

Course. | Title. | Date. Mo Tu We Th Fr Sa Su

Finding Unknowns:

Upper Four-Bar Loop: (NM-BJ)

$N\cos\theta_N + M\cos\theta_M - B\cos\theta_B - J\cos\theta_J = 0$

$N\sin\theta_N + M\sin\theta_M - B\sin\theta_N - J\sin\theta_N = 0$

$$\boxed{x\cos\theta_{\{J,B\}} + y\sin\theta_{\{J,B\}} \triangleq R\cos(\theta_{\{J,B\}} - \alpha) = A}$$

$x = N\cos\theta_N + M\cos\theta_M$
$y = N\sin\theta_N + M\cos\theta_M$

$R = \sqrt{x^2+y^2}$; $\alpha = \tan^{-1}(y/x)$

$$A_{\{J,B\}} = \frac{\{J,B\}^2 - \{B,J\}^2 + x^2 + y^2}{2\{J,B\}}$$

Lower Four-Bar Loop: (NM-CK)

$N\cos\theta_N + M\cos\theta_M - C\cos\theta_C - K\cos\theta_K = 0$

$N\sin\theta_N + M\sin\theta_M - C\sin\theta_C - K\sin\theta_K = 0$

$$\boxed{x\cos\theta_{\{K,C\}} + y\sin\theta_{\{K,C\}} \triangleq R\cos(\theta_{\{K,C\}} - \alpha) = A}$$

$x = N\cos\theta_N + M\cos\theta_M$
$y = N\sin\theta_N + M\cos\theta_M$

$R = \sqrt{x^2+y^2}$; $\alpha = \tan^{-1}(y/x)$

$$A_{\{K,C\}} = \frac{\{K,C\}^2 - \{C,K\}^2 + x^2 + y^2}{2\{K,C\}}$$

Parallel Loop: (DC-FG)

$D\cos\theta_D + C\cos\theta_C - F\cos\theta_F - G\cos\theta_G = 0$

$D\sin\theta_D + C\sin\theta_C - F\sin\theta_F - G\sin\theta_G = 0$

$$\boxed{x\cos\theta_{\{G,F\}} + y\sin\theta_{\{G,F\}} \triangleq R\cos(\theta_{\{G,F\}} - \alpha) = A}$$

$x = D\cos\theta_D + C\cos\theta_C$
$y = D\sin\theta_D + C\cos\theta_C$

$R = \sqrt{x^2+y^2}$; $\alpha = \tan^{-1}(y/x)$

$$A_{\{G,F\}} = \frac{\{G,F\}^2 - \{F,G\}^2 + x^2 + y^2}{2\{G,F\}}$$

Note: Link D is presumed to play the same role as Link N

Toe Loop: (CI-P)

$Pe^{i\theta_P} = Ce^{i\theta_C} + Ie^{i\theta_I}$

$\ln(e^{i\theta_P}) = \left(\frac{Ce^{i\theta_C} + Ie^{i\theta_I}}{P}\right)\ln$

$$\boxed{\theta_P = \ln\left(\frac{Ce^{i\theta_C} + Ie^{i\theta_I}}{P}\right)}$$

$C\cos\theta_C + I\cos\theta_I - P\cos\theta_P = 0$

$C\sin\theta_C + I\sin\theta_I - P\sin\theta_P = 0$

$x = C\cos\theta_C + I\cos\theta_I$
$y = C\sin\theta_C + I\sin\theta_I$

$$\boxed{P = \sqrt{x^2+y^2} \ ; \ \theta_P = \tan^{-1}(y/x)}$$

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su

Course.	Title.	Date.
		Mo Tu We Th Fr Sa Su